

TSENTRAL'NYY RUDNIK, Russian Federation

WMO: 296540

Lat: 55.206N

Lon: 87.642E

Elev: 498

StdP: 95.48

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.0	(c) -29.6	(d) -36.0	(e) 0.1	(f) -32.7	(g) -32.7	(h) 0.2	(i) -29.4	(j) 9.8	(k) -6.6	(l) 8.7	(m) -7.8	(n) 1.0	(o) 180	(p) 0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.1	(c) 27.7	(d) 17.4	(e) 25.7	(f) 16.4	(g) 24.0	(h) 15.8	(i) 18.6	(j) 24.8	(k) 17.7	(l) 23.3	(m) 16.9	(n) 22.1	(o) 2.5	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.6	20.5	15.7	11.8	19.8	14.8	11.1	19.1	55.0	24.8	52.1	23.6	49.5	22.1	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.4	7.4	6.4	-35.2	30.9	4.5	1.6	-38.4	32.1	-41.0	33.0	-43.5	33.9	-46.8	35.0
			-35.3	20.4	4.4	1.1	-38.5	21.3	-41.1	21.9	-43.6	22.6	-46.8	23.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.2	-15.0	-12.5	-5.5	1.6	8.4	15.3	16.9	15.0	8.4	1.3	-7.6	-12.7
	DBStd	12.57	7.58	6.60	6.07	5.62	5.43	4.26	3.42	3.70	4.76	5.61	7.18	6.84
	HDD10.0	3842	774	630	481	255	97	9	1	4	86	275	527	704
	HDD18.3	6304	1033	863	739	503	309	107	67	117	299	529	777	962
	CDD10.0	632	0	0	0	2	48	167	216	157	37	4	0	0
	CDD18.3	53	0	0	0	0	2	15	24	12	1	0	0	0
	CDH23.3	523	0	0	0	0	37	166	189	117	14	0	0	0
	CDH26.7	98	0	0	0	0	7	29	35	24	2	0	0	0
Wind	WSAvg	2.6	3.0	3.0	3.2	2.8	2.5	2.1	1.8	1.9	2.2	2.8	3.1	3.2
Precipitation	PrecAvg	989	55	47	66	62	78	100	93	102	93	109	98	87
	PrecMax	1333	128	110	284	137	187	194	161	233	177	171	163	290
	PrecMin	765	7	5	15	11	14	33	27	41	1	53	55	30
	PrecStd	155	29	22	53	28	42	43	35	42	41	30	31	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.9	2.0	11.5	18.0	27.4	29.5	30.0	29.7	25.3	18.2	9.0	1.1
		MCWB	-1.4	-1.1	4.3	8.2	15.4	17.4	18.8	18.6	14.5	9.6	4.6	-0.8
	2%	DB	-1.7	-0.3	7.7	15.1	23.5	27.2	27.3	26.5	21.3	14.9	5.7	-0.8
		MCWB	-2.8	-2.4	2.0	7.0	13.2	16.8	17.9	17.3	12.9	8.0	3.2	-1.8
	5%	DB	-3.8	-2.1	4.9	12.5	20.7	25.2	25.4	24.1	18.5	12.2	3.0	-2.7
		MCWB	-5.0	-3.5	0.6	5.7	11.4	15.8	17.1	16.2	11.8	6.8	0.8	-3.7
	10%	DB	-5.5	-3.7	2.7	10.1	17.9	23.2	23.5	21.9	15.8	9.2	0.7	-4.2
		MCWB	-6.5	-4.9	-0.7	4.7	10.5	15.1	16.4	15.4	10.7	5.5	-0.5	-5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.3	0.0	4.9	9.4	16.5	19.5	19.8	19.9	15.6	10.6	5.6	-0.3
		MCDB	0.7	0.8	10.6	15.8	25.5	26.1	26.0	27.6	22.9	15.7	7.3	0.7
	2%	WB	-2.7	-2.2	2.6	8.1	14.1	17.9	18.7	18.2	13.7	8.9	3.1	-1.7
		MCDB	-1.7	-0.8	6.4	13.5	20.7	24.4	25.0	23.8	19.7	13.3	5.5	-0.9
	5%	WB	-5.0	-3.5	0.9	6.5	12.7	16.9	18.0	17.1	12.6	7.4	1.2	-3.6
		MCDB	-4.1	-2.3	4.3	11.5	18.5	22.8	23.5	22.3	17.3	11.5	2.8	-2.8
	10%	WB	-6.5	-4.9	-0.4	5.0	11.2	16.0	17.2	16.1	11.3	5.8	-0.3	-5.1
		MCDB	-5.5	-3.6	2.3	9.5	16.8	21.6	22.1	21.1	15.0	8.9	0.6	-4.4
Mean Daily Temperature Range	5% DB	MDBR	7.2	8.1	9.0	9.2	10.3	10.1	9.1	9.5	8.5	6.7	6.0	6.3
		MCDBR	7.6	8.8	12.4	12.7	15.1	13.7	12.3	13.2	13.6	11.6	6.9	6.7
		MCWBR	6.7	7.3	7.9	7.1	7.4	5.6	5.1	5.7	6.5	6.3	4.7	6.0
	5% WB	MCDBR	6.9	7.9	10.7	11.4	12.6	11.3	10.3	11.0	11.4	10.2	6.7	6.3
		MCWBR	6.3	6.8	7.2	7.0	7.0	5.5	5.0	5.5	6.0	6.0	4.9	6.0
Clear-Sky Solar Irradiance	taub		0.293	0.333	0.383	0.412	0.418	0.410	0.433	0.415	0.378	0.363	0.331	0.299
	taud		1.976	1.932	1.854	1.954	2.144	2.277	2.218	2.239	2.282	2.160	1.971	1.918
	Ebn at Noon		578	693	750	798	823	837	806	794	773	667	512	455
	Edh at Noon		73	111	152	159	140	124	130	119	100	88	73	62
All-Sky Solar Radiation	RadAvg		0.87	1.86	3.19	4.32	4.99	5.83	5.28	4.33	2.86	1.47	0.72	0.54
	RadStd		0.10	0.15	0.31	0.30	0.60	0.51	0.37	0.34	0.36	0.17	0.12	0.09

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page